

# Byeongjee Kang

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## Research Interests

My research is on language and compiler support for automatically generating safe and efficient programs. I combine static analysis with compiler infrastructure built on LLVM and MLIR. My recent interests are in resource-constrained platforms, where programs require checkpointing or migration to safely execute under strict energy and memory budgets.

## Technical Skills

**Expertise:** Compiler engineering, static analysis, information flow analysis, embedded systems, formal verification

**Languages & Tools:** Rust, C++, Python, SML, Maude, Futhark, WebAssembly, LLVM, MLIR, rustc (MIR)

## Education

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| <b>PhD</b> | <b>Carnegie Mellon University</b> , Software Engineering<br>Advised by <a href="#">Prof. Limin Jia</a>                                                                                                                                                                                                                                                                                                                                                                                                       | Pittsburgh, USA<br>Aug 2023 – May 2028<br>(expected) |
| <b>BS</b>  | <b>Pohang University of Science and Technology (POSTECH)</b> , Computer Science and Engineering & Mathematics (double major) <ul style="list-style-type: none"> <li>GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023)</li> <li>Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA</li> <li>Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)</li> </ul> | Pohang, Korea<br>Feb 2018 – Feb 2023                 |

## Work Experiences

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| <b>Microsoft Research</b> , Research intern at Data Systems Group mentored by Kaushik Rajan <ul style="list-style-type: none"> <li>Built auto-vectorizing compiler optimization passes using MLIR and Polygeist, vectorizing loops with control-flow dependencies that are not supported by existing loop vectorizers</li> </ul>                                                                                                                                                                           | Redmond, WA, USA<br>May 2025 – Aug 2025 |
| <b>Codebox</b> , Software engineer as military service alternative <ul style="list-style-type: none"> <li>Developed the consensus engine of <a href="#">CodeChain Foundry</a>, an open-source blockchain engine with a composable module system, implementing BLS signature aggregation for cryptographic vote verification and the interface between modules and the consensus engine</li> <li>Built <a href="#">Zuzu</a>, a web-based cap table management software, as a full-stack engineer</li> </ul> | Seoul, Korea<br>Jan 2020 – Dec 2021     |

## Publications

### Bao: Automatic Region Placement and Memory Allocation for Intermittent Computing

*LLVM-based compiler that optimizes checkpoint placement and memory allocation for intermittently powered devices by solving a mixed-integer linear program*

**Byeongjee Kang**, Jan Hoffmann, Limin Jia, Brandon Lucia, Feras Saad

(International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2027 (to appear)) [\[code\]](#)

### WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction

Aug 2026

*MLIR-based compiler from high-level languages to WebAssembly that takes advantage of high-level Wasm features such as stack switching*

**Byeongjee Kang**, Harsh Desai, Limin Jia, Brandon Lucia

[10.1145/3828170.3828172](#) (International Conference on Managed Programming Languages and Runtimes (MPLR) 2026) [\[code\]](#)

### **Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM**

Oct 2025

Ao Li, **Byeongjee Kang**, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye

[10.1145/3764119](#) (Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) 2025) [\[code\]](#)

### **GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits**

Sept 2024

Sam Westrick, Pengyu Liu, **Byeongjee Kang**, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar

[ieeexplore.ieee.org/document/10821338](https://ieeexplore.ieee.org/document/10821338) (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper))

### **Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems**

July 2024

*Refutation-complete symbolic model checking of distributed systems in Maude using narrowing with heuristic search*

**Byeongjee Kang**, Kyungmin Bae

[10.1016/j.scico.2024.103097](#) (Science of Computer Programming 2024) [\[code\]](#)

### **Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search**

Dec 2022

**Byeongjee Kang**, Kyungmin Bae

[dl.acm.org/doi/10.1145/3563822.3568017](https://dl.acm.org/doi/10.1145/3563822.3568017) (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022) [\[code\]](#)

## **Software**

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- **Bao**: LLVM toolchain that instruments programs for batteryless devices by solving a mixed-integer linear program
- **WAMI**: MLIR-based compilation pipeline for WebAssembly
- **OCaMI** (work in progress): Compiler from a functional subset of OCaml to WebAssembly through WAMI
- **rustlane** (work in progress): ISPC-style SPMD programming for Rust, lowering scalar control flow to masked SIMD
- **Fray**: Controlled concurrency testing framework for the JVM
- **heuristic-narrowing-search**: Maude implementation of narrowing-based symbolic reachability analysis
- **Simperby**: BFT blockchain engine for decentralized autonomous organizations (DAOs). Designed **Vetomint**, its consensus protocol, which extends Tendermint with leader veto and early termination
- **CodeChain Foundry**: Open-source blockchain engine based on composable module system

## **Ongoing Research Projects**

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- **Language-Level Program Migration**: Compiler and runtime support for migrating live Rust programs under strict time and energy budgets by erasing reinitializable state at the source and rebuilding it at the destination
- **Probabilistic Energy Estimation**: Probabilistic modeling and analysis tools for estimating energy consumption of programs on resource-constrained devices
- **Cumulative Memory Analysis**: Type-based automatic amortized resource analysis that infers symbolic bounds on memory usage integrated over execution time

## **Honors and Scholarships**

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### **ILJU Fellowship (ILJU Academy and Culture Foundation)**

Aug 2023 – July 2028

- Received total of USD 120,000 during graduate studies

### **Global Leadership Program (CSE Department, POSTECH)**

Sept 2019 – Feb 2023

- Scholarship and research support program for selected students in the CS department

**Domun Scholarship (POSTECH)**

Feb 2019

- Scholarship for freshmen with the highest GPA

**Program for Highly Dedicated Students (POSTECH)**

Feb 2018 – Feb 2023

- Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year

**Presidential Science Scholarship (Ministry of Education of Korea)**

Feb 2018 – Feb 2023

- National scholarship offered to top freshmen students in the field of science and technology

**Talks**

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- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [\[recording\]](#)